

Basics of Linux OS

1. Linux Prompt

In Linux, a prompt is what you see in the terminal before you type a command. It provides key information about your current shell session.

Example:

```
[root@server ~]#  
[user@server ~]$
```

Breakdown:

- **root** → Current logged-in user.
 - **root** is the superuser or administrator account with full privileges to control the entire system.
 - **server** → The hostname of the machine. Used to identify the computer within a network.
 - **~** (tilde) → Current working directory.
 - **~** means the user's home directory.
 - **#** (hash) → Indicates the user is the root (administrator) and has full privileges.
 - **\$** (dollar) → Indicates the user is a regular user with limited privileges. To execute admin-level commands, you usually use **sudo**.
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2. Basic Linux Commands

Here are some essential Linux commands and their usage:

2.1 User and System Info

- **whoami** → Displays the current logged-in username.
- **hostname** → Displays the system's machine (host) name.

- `hostnamectl` → Shows detailed hostname and system information. Can be used to set system name.
 - `exec bash` → Reloads the current shell, applying updated configurations without restarting the terminal.
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2.2 Files and Directories Management

- `ls` → Lists files and directories in the current directory.
 - Options: `-l` for detailed list, `-a` for hidden files.
 - `touch file.txt` → Creates a new empty file or updates timestamp of existing file.
 - `mkdir dir_name` → Creates a new directory.
 - Use `-p` to create parent directories if they don't exist.
 - `cat file.txt` → Displays file content; can also combine multiple file contents.
 - `history` → Shows previously executed commands.
 - Use `history -c` to clear history.
 - `clear` → Clears terminal screen.
 - `date` → Displays or sets system date and time.
 - `cal` → Displays a calendar.
 - Example: `cal 8 2025` (August 2025)
 - `head file.txt` → Displays the first 10 lines of a file. Use `-n` to specify a number.
 - `tail file.txt` → Displays the last 10 lines of a file. Use `-n` to specify a number.
 - `uname` → Shows system information. Use `-a` for all details.
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2.3 Disk and Memory Utilities

- `free` → Displays RAM and swap memory usage.
 - `du` → Shows disk usage of files and directories.
 - `df` → Shows free disk space. Use `-h` for human-readable sizes.
 - `file filename` → Identifies the file type.
 - `tree` → Shows directory structure in a tree-like format.
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2.4 File and Directory Deletion

- `rm filename` → Deletes a file.
- `rmdir dir_name` → Deletes an empty directory.
- `rm -rf dir_name` → Deletes a directory and its contents forcefully.

- `-r` → Recursive delete
 - `-f` → Force delete without prompt
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2.5 Command Options Syntax

`command [options] [arguments]`

Example:

```
ls -l /home/user
```

- command → `ls`
 - option → `-l` (detailed list)
 - argument → `/home/user` (directory to list)
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3. Command History Shortcuts

- Up Arrow → Browse previous commands one by one.
 - `!
<number>` → Re-run a specific command from history based on ID number.
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4. Linux Editors

Popular editors:

- gedit → GUI-based text editor.
 - vi / vim → Command-line text editors.
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4.1 Modes in vim

- Insert Mode → Used for typing text.
 - Enter insert mode: `i`, `a`, `o`
- Command Mode → Used for executing editor commands. Enter by pressing `Esc`.
- Execute Mode → Directly issues commands like save, quit, replace.

4.2 Useful vim Commands

In Command mode:

- `:set number` → Show line numbers.
- `:set nonumber` → Hide line numbers.
- `:/pattern` → Search for a string.
- `:noh` → Remove search highlight.
- `:w` → Save file.
- `:q` → Quit without saving.
- `:wq` → Save and quit.
- `:saveas new_name` → Save file as new name.
- `:%s/pattern/replace/g` → Search and replace all occurrences.

In Execute/Normal mode:

- `yy` → Copy line.
 - `p` → Paste.
 - `nyy` → Copy n lines.
 - `yiw` → Copy a word.
 - `dd` → Delete line.
 - `ndd` → Delete n lines.
 - `x` → Delete character.
 - `u` → Undo.
 - `G` → Go to end of file.
 - `gg` → Go to top of file.
 - `n+G` → Jump to nth line from bottom.
 - `n+gg` → Jump to nth line from top.
 - `zz` → Save and quit.
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5. Linux File Hierarchy

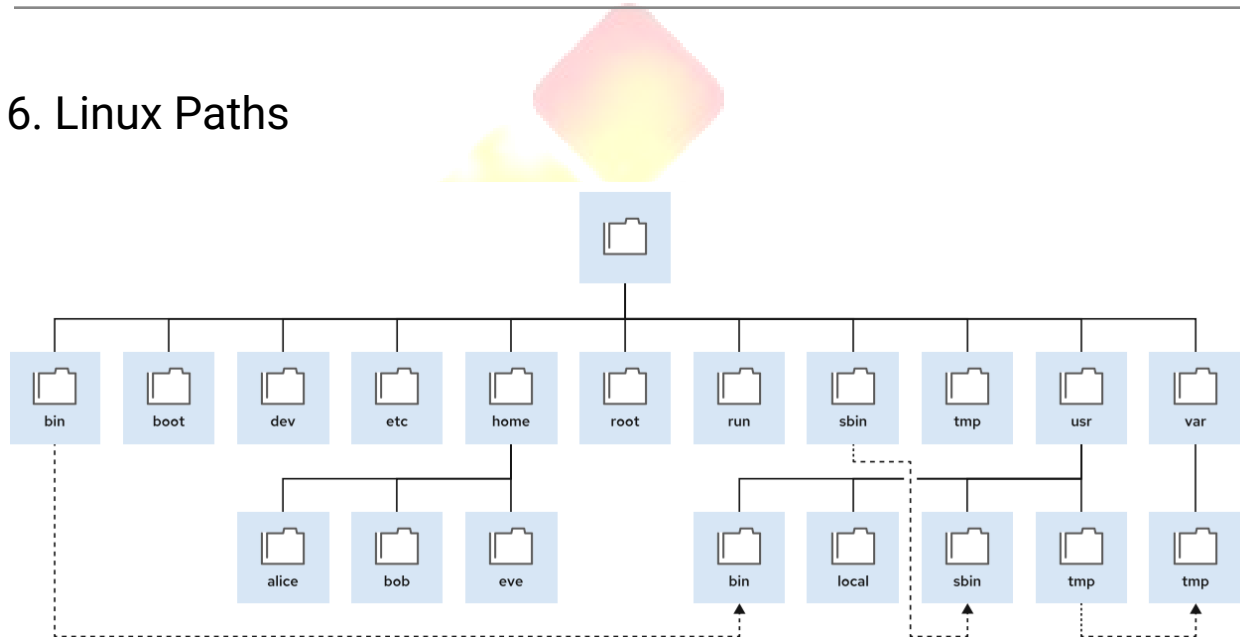
Linux directories are organized as per Filesystem Hierarchy Standard (FHS).

Main Directories:

- `/` → Root directory, top of hierarchy.

- `/bin` → Essential command binaries.
- `/sbin` → System binaries (admin commands).
- `/boot` → Boot loader and kernel files.
- `/dev` → Device files for hardware.
- `/etc` → Configuration files.
- `/root` → Home directory of `root`.
- `/home` → User home directories.
- `/mnt` → Temporary mount points.
- `/tmp` → Temporary files.
- `/srv` → Data for system services.
- `/usr` → User programs and libraries.
- `/var` → Variable data (logs, mail, database).
- `/lib` → Shared libraries.
- `/proc` → Process and kernel info (virtual filesystem).

6. Linux Paths



Types of Paths:

- Absolute Path → Starts with `/`. Full path from root.
 - Example: `/home/user/file.txt`
- Relative Path → Relative to current directory.
 - Example: `docs/file.txt`

Special Symbols:

- `.` → Current directory.
 - `..` → Parent directory.
 - `~` → Current user's home.
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7. Copy, Move, Rename

Copy (cp)

`cp [options] source destination`

Examples:

- Single file: `cp file.txt /path/`
- Multiple files: `cp file1 file2 /path/`
- Directory: `cp -r dir /path/`

Options:

- `-v` → Verbose
 - `-f` → Force overwrite
 - `-p` → Preserve attributes
 - `-n` → No overwrite
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Move / Rename (mv)

`mv [options] source destination`

Examples:

- Move file: `mv file.txt /path/`
 - Rename file: `mv old.txt new.txt`
 - Move directory: `mv dir /path/`
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8. Viewing Large Files

- `more file.txt` → View file one page at a time (limited backward navigation).

- `less file.txt` → View file interactively (scroll up/down, search).
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